Simulating the False Vacuum via Quantised Interacting Bubbles.

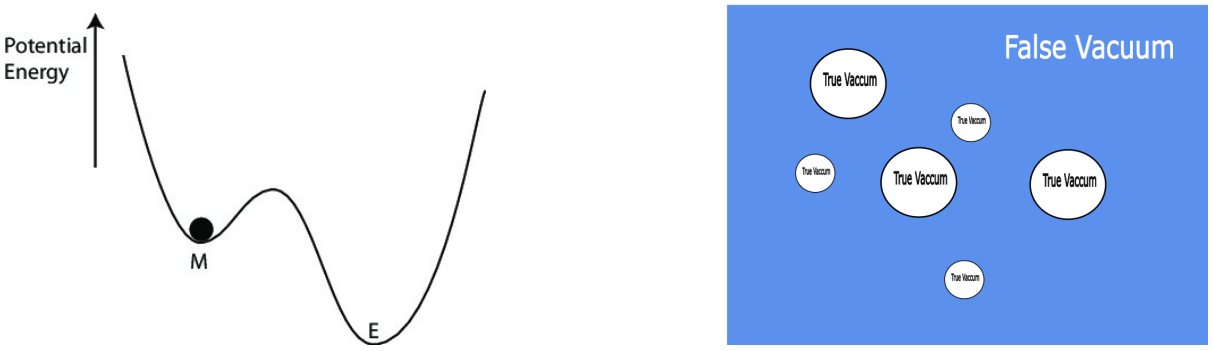
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1 Background.

Major breakthroughs of modern quantum field theories [1] have shown that many systems thought to be classically stable are actually in a metastable energy state due [2, 3] to quantum fluctuations [4]. As represented in fig. 1a, a metastable state is one where the object of interest is in a 'well', where, despite not being at the lowest point (the global minimum), it does not have the energy required to overcome the surrounding barriers, so remains in its "metastable" state. Quantum Mechanics proposes the concept of quantum tunnelling [5], whereby particles can overcome seemingly impossible potential energy barriers, due to their probabilistic wave nature [6]. In the analogy provided in fig. 1a imagine the ball going *through* the hill instead of over it. Excitation of low energy states, originally believed to be caused by thermodynamic phenomena, was instead revealed to be ground-state excitations of the quantum field [7, 8], and advances in theoretical particle physics have further implied that the electro-weak vacuum itself could also be metastable [9–18]. This meant that, even in a vacuum, there exist small fluctuations of energy at a quantum scale [19]. It was then proposed that [20], through quantum tunnelling, small, interacting bubbles of a true vacuum could form (fig. 1b). So called "False-Vacuum Decay" uprooted classical physics, as a transition from a metastable energy state to a lower, ground state would alter the fundamental constants used in physics. [21]

The false vacuum decay process is of a non-perturbative nature [7], meaning that attempts to analytically represent the theoretical framework of such a process are limited. The tunnelling process itself is invisible to any perturbative calculation [22] (one that starts with a classical system, then makes approximations accordingly), and perturbation theory can only be applied to small fluctuations in stable systems [23], unlike the large fluctuations of the quantum field (known as instantons [24],) used to represent field decay. This, altogether, means that the probability of a tunnelling process occurring is near zero. Furthermore, dynamics of quantum field interactions take *immense* computational power to simulate [25]. Our best option, therefore, is to represent the process in Quantum Many Body systems. Examples of such systems include ferromagnetic superfluids [26, 27], where bubbles are created inside bosonic gases allowing us to study interactions of the field through the bubbles. More common methods involve either neutral atom systems [28, 29] or the Quantum Ising model [2, 3, 30–39], which will be the focus of this paper. These allow us to simulate and probe interactions between bubbles of true



(a) (b)

Figure 1: (a) demonstrates a ball in a local minimum, M, with a nearby global minimum, E. This diagram is taken from [53], but is common used. (b) shows a visualisation of the "bubble" aspects of true vacuum (in various "quantised" sizes) expanding into the false vacuum.

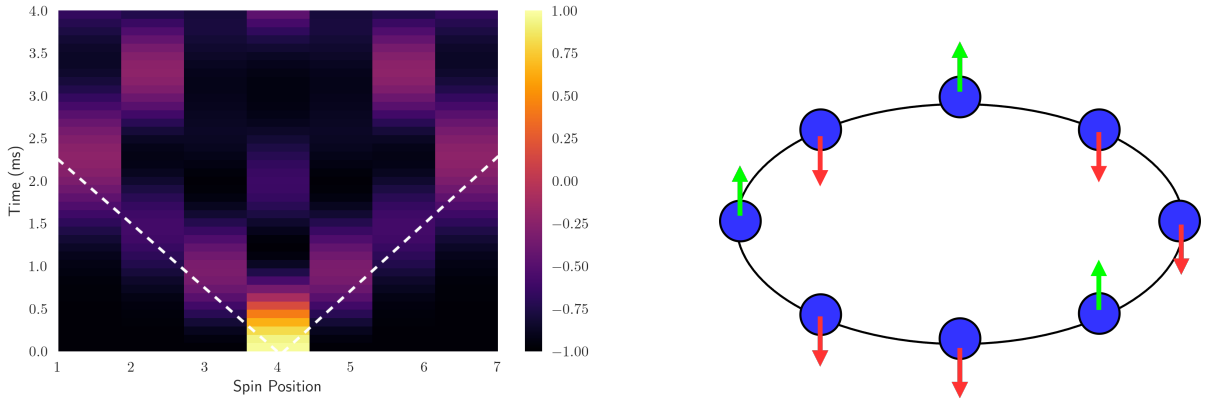
vacuum in high energy systems, via chains or rings of entangled particles.

More recent experiments have involved systems with upwards of thousands of interacting qubits [30]. However, a major limitation of using so many qubits is decoherence [40]: as more and more qubits are involved, the consideration toward interactions between qubits and their environment becomes larger. A specific type of quantum computer called a quantum annealer [41] must then be used to solve optimisation problems to calculate the global minimum of the system. Compared to a gate-based computation [42], where qubits need to be kept fully isolated from their environment, the annealer needs only to begin the system in an easily computable state, and evolves the system adiabatically such that it remains in its ground state.

Research into quantum annealing and machine learning is itself a highly impactful area of research in quantum information science [43–46], although this is outside of the scope of this project.

Since its initial outlining by Coleman [7, 8], False Vacuum Decay has provided a prominent area of research across condensed matter, cosmology [47, 48] and quantum field theory. Analytic limitations have advanced our technology and understanding of physical scenarios, like quantum phase transitions [49, 50], spin glasses [51], and the Kibble-Zurek Mechanism [52].

There are two main aims of this project. First, we wish to identify and describe various areas of research surrounding false vacuum decay and the quantum Ising model, identifying techniques such as Lieb-Robinson cones and magnetisation through quench dynamics. Secondly, we wish to provide, in more detail, an explanation of the methods used to simulate the decay process itself. We therefore will be able to provide the optimal parameters to observe the decay in different scenarios. Consideration will also be given to how these effects are altered in a real-world scenario, Cobalt Niobate, and how through experimental research into this Ising-like model, the underlying theory behind these models can be advanced. Throughout this dissertation, all figures and diagrams have been produced by the author unless otherwise stated.



(a) (b)

Figure 2: (a) Lieb-Robinson light cone for a 1D Ising model of $N = 7$ spins with periodic boundary conditions, at $J = 1$, $h_z = 3$. An initial spin flip occurs at the fourth spin at $t = 0$. The system is then left to evolve over time, and the observed light cone, demonstrated by the dashed white line, demonstrates the speed at which spin flips propagate through the system. The colour scale represents the expectation value of the local magnetisation, $\langle \hat{\sigma}_j^z \rangle$ of each spin at different times t. This graph is similar to that of [54], however note slight differences arising from applying periodic boundary conditions and different initial parameters. Our graph is also of an idealised scenario, most notably in the absence environmental effects. (b) representation of the quantum Ising model with periodic boundary conditions, for the state $|\uparrow\downarrow\downarrow\uparrow\downarrow\downarrow\uparrow\downarrow\rangle$

2 The Ising Model.

The most common method used in simulations is the Quantum Ising Model (fig. 2b). This is for a multitude of reasons [55], most notably because a Jordan-Wigner transformation [56] will exactly map the provided spin-model onto a fermionic theory [57]. The reason that the Ising model accurately represents a quantum field is because near its critical point (the configuration such that it undergoes a phase transition from ferromagnetic to paramagnetic), the model shares the same symmetries as a free fermion quantum field theory in the continuum limit. The Ising model is one of very few configurations in statistical field theory that can be solved exactly [58], leading to its rigorous use in quantum and conformal field theory [59].

The one-dimensional model itself [60], is a lattice structure of spin $\pm\frac{1}{2}$ particles, arranged in a chain or ring, such that each particle only feels the effect of its nearest neighbour. The Hamiltonian that governs a system of N particles is given as :

$$\hat{H} = -J \sum_{j=1}^{N} \hat{\sigma}_j^z \hat{\sigma}_{j+1}^z - h_x \sum_{j=1}^{N} \hat{\sigma}_j^x \tag{1}$$

where we use the notation:

$$\hat{\sigma}_i^z = \mathbb{I} \otimes \mathbb{I} \otimes ... \otimes \sigma_z \otimes \mathbb{I} \otimes ... \otimes \mathbb{I}$$

with σ_z being a Pauli matrix in the i-th position. The $-J \sum_{j=1}^{N} \hat{\sigma}_j^z \hat{\sigma}_{j+1}^z$ term describes the pairing of particles in the model's spin states with that of its nearest neighbour. This term

means that a more favourable (i.e lower) energy state will occur when neighbouring spins are aligned. The h_x component induces disorder into the system by generating quantum fluctuations, allowing it to more accurately represent quantum dynamics. These terms are summed over every "site" within the model. For the ring Ising model, we employ a boundary condition of $\hat{\sigma}^{\alpha}_{N+1} = \hat{\sigma}^{\alpha}_{1}$, to ensure that all interactions are accounted for.

The system has a $\mathbb{Z}_2$ symmetry [61], meaning that if every spin within the system is flipped, the state of the system remains unchanged. This intuitively leads to a doubly degenerate ground state, where both $|\uparrow\uparrow\uparrow ... \uparrow\rangle$ and $|\downarrow\downarrow\downarrow ... \downarrow\rangle$ are ground states with equal energies. This model can be used to represent a system with both a false and true vacuum state. However, to accurately simulate the dynamics of false vacuum decay, we need to break this symmetry so that one "ground" state is at a lower energy minima than the other.

If a system is prepared in the ground-state, $|GS\rangle$, of this Hamiltonian, flipping the central spin via $|\psi_0\rangle = \hat{\sigma}^{x}_{\frac{N+1}{2}}|GS\rangle$ will induce an excitation wave that will propagate through the system via spin flips, with a finite speed, v_{LR}. Since spin is effectively the net magnetic moment, we can gather from the magnetisation the most likely axis for the spin to be aligned with. The speed of this propagation is known as the Lieb-Robinson velocity [62]. Using our code based on [54], we were able to produce fig. 2a, which demonstrates this effect in a chain of 7 $\downarrow$ spins, with the central (fourth) spin being flipped at time $t = 0$. The observed light cone indicates that spins closer to the the initial perturbation are affected more immediately than those further away. This demonstrates a fundamental aspect of quantum dynamics known as the Lieb-Robinson bound, describing the theoretical upper limit at which information in a non-relativistic system can propagate [63]. Now that we have an understanding of how this system interacts with itself, we can perturb the system (for example through applied magnetic fields) in order to achieve false vacuum decay.

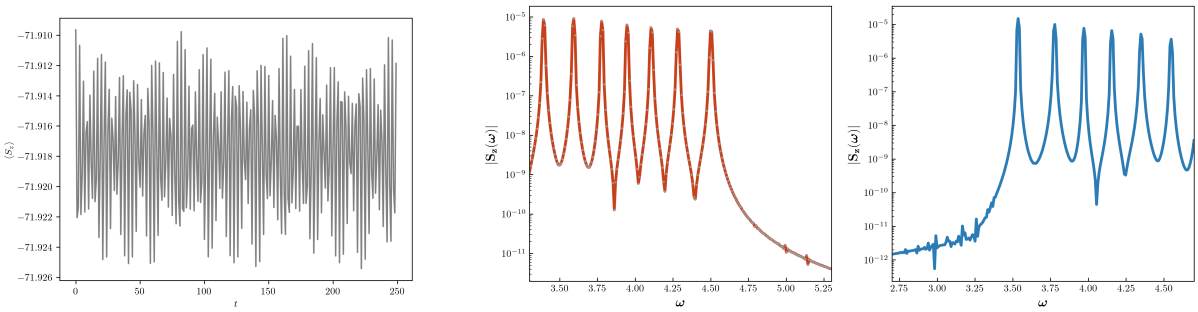
3 Magnetisation and Quench Dynamics.

When a transverse magnetic field is applied, the Hamiltonian of the Ising model becomes:

$$\hat{H} = -J\sum_{j=1}^{N}\hat{\sigma}^{z}_{j}\hat{\sigma}^{z}_{j+1} - h_x\sum_{j=1}^{N}\hat{\sigma}^{x}_{j} - h_z\sum_{j=1}^{N}\hat{\sigma}^{z}_{j} \tag{2}$$

known as the "Transverse Ising Model" [64]. The new h_z term is used to impose an energy bias between the two ground states. This is because, with the inclusion of the transverse magnetic field, the system has lost its $\mathbb{Z}_2$ symmetry, and the $|\uparrow\uparrow\uparrow ... \uparrow\rangle$ and $|\downarrow\downarrow\downarrow ... \downarrow\rangle$ no longer have the same energy. This provides us with the necessary framework to represent false vacuum decay, as so long as $h_z > 0$, the state $|\psi_{\uparrow}\rangle = |\uparrow\uparrow\uparrow ... \uparrow\rangle$ is representative of the true vacuum state and $|\psi_{\downarrow}\rangle = |\downarrow\downarrow\downarrow ... \downarrow\rangle$ that of a false vacuum.

With the Hamiltonian constructed, we are then able to begin simulating the false vacuum decay process. We proceed by identifying the smallest algebraic eigenstate of $\hat{H}$ and calculate



(a) (b)

Figure 3: Evolution of the $|\downarrow\downarrow\downarrow ... \downarrow\rangle$ ground state of a 12 spin system with $\hat{H}$ such that $J = 1$, $h_x = 0.1$ and $h_z = 0$. The magnitude of $h_x = 0.1$ was chosen to induce light, quantum dynamics into the system as will often be the choice in this paper. The wavefunction is time evolved for an arbitrary time until it (roughly) converges to an eigenstate. A transverse field $h_z = 0.1$ is applied and the system is unitarily evolved for 250 time units. We take the expectation value $\langle S_z(t)\rangle$ after each increment of $dt = 1$. (a) demonstrates the oscillations between negative magnetisation values. (b) and (c) show the Fourier transform at $h_x = 0.2$ and $h_z = \pm 0.04$ for a false vacuum (Orange) and a true vacuum (blue). These figures are similar to that of [39], but for a smaller system size.

its associated energy. This will be the lowest energy of the system and therefore the ground state. We then time evolve this eigenstate according to the usual equation:

$$|\Psi\rangle = e^{-i\hat{H}t}\,|\downarrow\downarrow\downarrow ... \downarrow\rangle \tag{3}$$

(where we have set $\hbar = 1$, in accordance with natural units [65].) This process is repeated across ranging values of h_z in order to later determine the ideal resonance at which different sized bubbles of true vacuum will form.

Once the system is orientated in this state, we perform a quantum quench [66], by non-adiabatically flipping the sign of h_z, inducing a first-order quantum phase transition [67] that will swap the true and false vacua. The system is now orientated in the false vacuum state, with a desire to decay into the true vacuum state. The Hamiltonian is then left to evolve over time, and the subsequent dynamics can be observed.

In order to calculate the magnetisation, we instead need to evolve the ground state of the Hamiltonian at $h_z = 0$ through imaginary time evolution [68]:

$$|\psi(\tau)\rangle = e^{-\tau\hat{H}}\,|\psi_0\rangle \tag{4}$$

repeatedly for small increments of τ, normalising after each increment, until the system roughly converges to an eigenstate of $\hat{H}$, at which point:

$$\delta\hat{H} = \langle\psi(\tau)|\hat{H}^2|\psi(\tau)\rangle - (\langle\psi(\tau)|\hat{H}|\psi(\tau)\rangle)^2 \approx 0 \tag{5}$$

The obtained state is one of the degenerate ground states of the Ising model at $h_z = 0$, and in

this state the magnetisation will be negative. We then apply the transverse field to split the degenerate states and unitarily time evolve this wave function with the new Hamiltonian. As this time evolution occurs, we take the magnetisation as:

$$\langle S_z(t)\rangle = \frac{1}{2N}\sum_i \langle \Psi(t)|\hat{\sigma}_i^z|\Psi(t)\rangle \tag{6}$$

where $|\Psi(t)\rangle$ is the unitarily evolved state at time t. During this time evolution, the value of magnetisation will oscillate between small values in the negative region as in fig. 3a. By then taking a Fourier transform to the frequency domain for the appropriate region of frequencies [39], via:

$$|S_z(\omega)| = \left|\frac{1}{T}\int_0^T \langle S_z(t)\rangle\, e^{i\omega t}\mathrm{d}t\right| \tag{7}$$

we can represent the magnitude spectrum of magnetisation as the system is unitarily time evolved at $h_z > 0$ (true vacuum), and $h_z < 0$ (false vacuum). We expect to observe sharp peaks corresponding to the discrete energy levels of the system caused by the energy limitations imposed on the bubbles. Furthermore, we expect the peaks of a false vacuum to be more tightly spaced, representative of the fact that the false vacuum is a much more chaotic configuration, with more allowed energy levels. By programming eq. (6) and then taking the Fourier transform via eq. (7) we were able to produce fig. 3b. These demonstrate the results of time evolution for $h_z = 0.04$ (blue) and $h_z = -0.04$ (orange). From these results we can determine the scale of the energy gaps (distance between peaks) of the system. Each peak corresponds to an energy gap

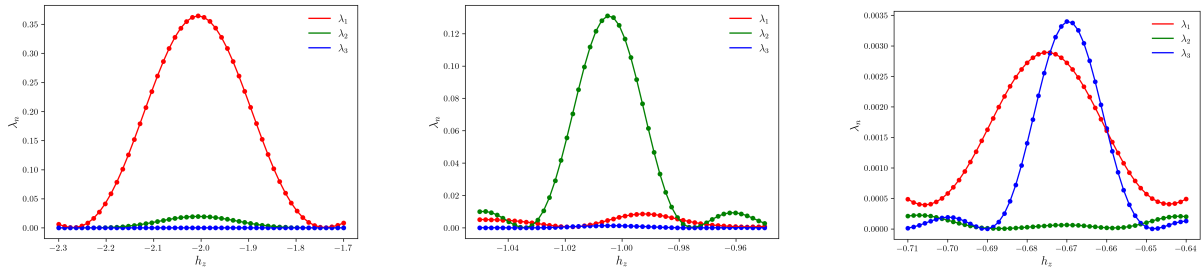
$$\omega = E_n - E_m \tag{8}$$

for two separate energy levels E_n, E_m. We also observe an accurate representation of false and true vacuum dynamics, and show that the two different vacua have largely different frequency oscillations, which is to be expected since they occur at different phases of the model itself. Now that the system has been properly initialised, we can begin to simulate the expected densities of true vacuum bubbles that will appear for ranging values of h_z.

4 Detecting the True Vacuum.

Theoretical prediction by Coleman [7] denotes that the size of a formed bubble of true vacuum will be dependent on the proportion of internal energy volume ($4J$), compared to surface tension in the domain boundary between false and true vacuum ($2nh_z$). The underlying theory therefore denotes that small bubbles that form would quickly shrink to nothing. However, if a particularly large bubble were formed, it would be rapidly expansive, and could convert the entire false vacuum into true vacuum (hence, false vacuum decay).

In the Ising model, bubbles of true vacuum are represented by chains of spins aligned opposite to the ground state (in our case, spins aligned downwards). Domain walls are then represented by the neighbouring up spins. For an n-bubble (one consisting of n many flipped spins), the



(a) (b) (c)

Figure 4: Bubble densities in an N= 12 Ising model with periodic boundary conditions. We set $J = 1$ to simplify the system (as we are not consider any specific material), and the subsequent graph displays the value of λ_n across ranging values of h_z. We set $h_x = 0.05$ for (a) because in this smaller regime, the one-bubble will be most prominent because of its lower resonance. Conversely, we set $h_x = 0.1$ for (b) and (c) to amplify the resonance of larger n bubbles, allowing 2 and 3 bubbles to be detected. The system is allowed to evolve for 10 time units (a) and 50 time units (b) and (c). The time for (b) and (c) was chosen to allow the smaller bubbles more time to nucleate; whereas, since one-bubbles are more commonly detected we limit the time evolution to avoid thermalisation. λ_1,Ψ,h_x

bubble density at a given time t, $\lambda_n(t)$ is calculated via [30]:

$$\lambda_n = (\frac{1}{N}) \sum_{i=1}^{N} \langle \hat{P}_i^{\uparrow} [\prod_{j=1}^{n} \hat{P}_{i+j}^{\downarrow}] \hat{P}_{i+n+1}^{\uparrow} \rangle \tag{9}$$

here $\hat{P}^{\phi} = |\phi\rangle\langle\phi|$ are projectors for some arbitrary state $|\phi\rangle$.

For example, when working in the basis

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

the up projector, $\hat{P}^{\uparrow}$ is defined as:

$$\hat{P}^{\uparrow} = |\uparrow\rangle \langle\uparrow| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

These projectors are used to detect when spin states are in the excited state, as $\hat{P}^{\downarrow}$ will return a 1 when applied to an $\downarrow$ state and a zero otherwise. The operator $\hat{P}^{\uparrow}\hat{P}^{\downarrow}\hat{P}^{\uparrow}$ will therefore detect a one-bubble ($\uparrow\downarrow\uparrow$) within a site, whilst the operator $\hat{P}^{\uparrow}\hat{P}^{\downarrow}\hat{P}^{\downarrow}\hat{P}^{\uparrow}$ detects a two-bubble, etc. λ_n applies this to each site within the given Ising model, using the same boundary conditions as in eq. (1), and calculates the expectation value of each n-bubble for a given wavefunction. We study the dynamics of n = 1, 2 and 3 bubbles and their associated density over ranging initial parameters. It has been shown in Vodeb et al [30], that bubbles for $n \leq 6$ will have the highest chance of detection according to $h_z = -2J/n$, with bubbles scarcely detected elsewhere. These peaks occur at such value because the energy gained by the bubbles becomes equal to the loss of energy caused by spin flips, whereas elsewhere energy restrictions means it is simply not possible for bubbles to form.

Based on the parameters outlined by Lagnese et al [33], we determined the ideal parameters to use for our system. By programming eq. (9) we were able to produce figs. 4a to 4c which show the values of λ_n at $h_z = 2, 1$ and $\frac{2}{3}$, and we show that detection of true vacuum is in fact possible in this regime. We set a larger h_x for larger bubbles, as they require a more disordered system. Consistent with the initial outline by Coleman, [7], we indeed observe that quantised bubbles are able to form. A peak in λ_1 is observed at $h_z = -2$. Similarly, for (b), we observe a smaller peak near the expected value of $h_z = -1$. Finally, a very small peak in λ_3 is observed when $h_z \approx -0.667$, however in this smaller region oscillations in λ_1 are also common. This is because fluctuations in single-spins will never fully subside, and essentially become part of the background noise, which is consistent with "stage 1" of Maertens et al [2] which suggested that in the first "stage" of false vacuum decay quantum fluctuations would drive the nucleation of smaller bubbles. In order to detect larger bubbles, it would be necessary to further reduce the value of h_z, and induce a more chaotic system. However, in an $N = 12$ site system this would likely cause thermalisation of the system. This section has therefore shown that it is possible to detect bubbles of true vacuum using a programme of the Ising model, and from here the obvious next step is to consider the state of these systems after initialisation.

5 Bubble Interactions.

With the peaks detected, we are now able to begin simulating how the bubbles evolve over time. First, it is necessary to remove the applied transverse magnetic field. This is because, since the degenerate vacua has been achieved we can observe the isolated bubble interactions without any bias-driven expansion. We start with a state comprising of only one, two or three bubbles, and apply a small h_x field, introducing quantum dynamics into the system and allowing bubbles to move, expand and collide via spin flips. We then time evolve these states according to the usual time evolution (eq. (3)).

We repeat this time evolution for small time increments, and calculate the expectation value of λ_n for the evolved state $|\psi(t)\rangle$ at each increment. By programming the desired initial state and then time evolving in increments via eq. (3), we were able to produce concrete results in figs. 5a to 5c. We observe that in each case, after a short amount of time bubbles of new sizes not initially seen in the ground state are expected. However, when this occurs, the expectation value of the more dominant bubble decreases accordingly. This is suggestive of Coleman's [7] prediction that a bubble is not able to expand on its own, and only by "absorbing" another bubble is it able to grow in size. Further, this aligns with the results of Pomponio et al [3] who used Bloch oscillations to represent the process, and found difficulty in observing the expansion of bubbles.

As previously referenced, Maertens et al [2] split the decay process into three distinct "stages". In stage two, it is believed that the observation of small bubbles would mostly halt, and larger bubbles would become dominant. Then, in stage 3, Bloch oscillations would impose a limit on

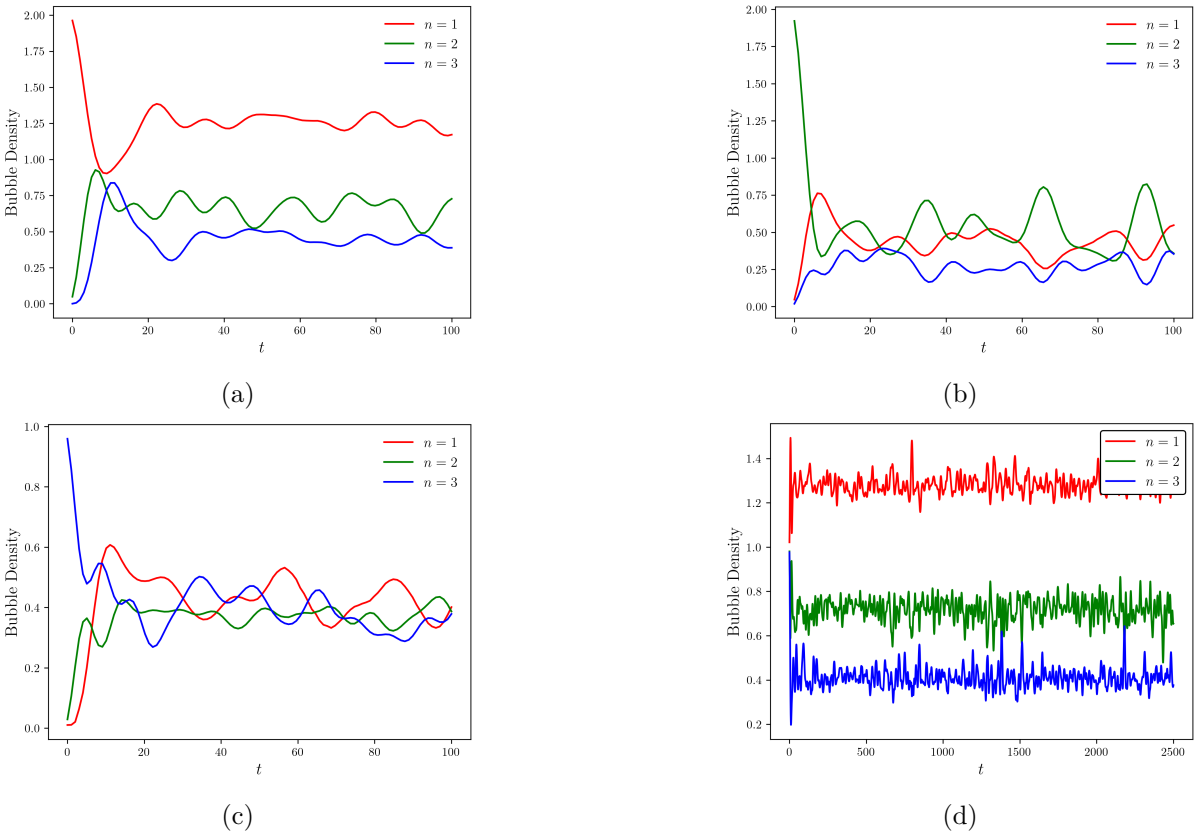
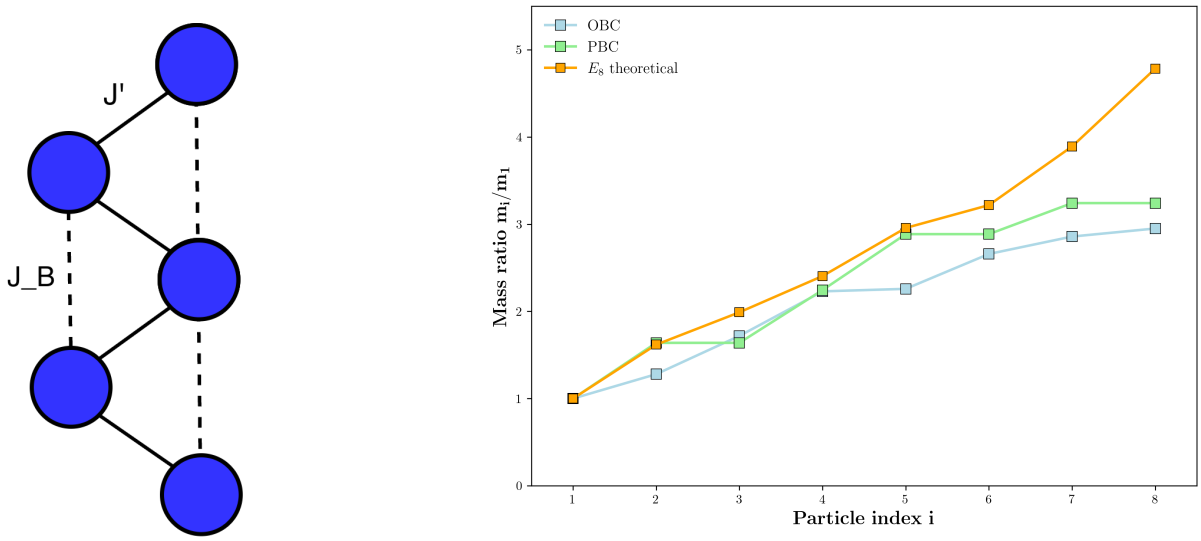


Figure 5: Unitary time evolution of the mixed states $|\uparrow\uparrow\uparrow\downarrow\uparrow\uparrow\downarrow\uparrow\uparrow\uparrow\downarrow\uparrow\rangle$ (a), $|\downarrow\downarrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\rangle$ (b), $|\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow\downarrow\uparrow\rangle$ (c), and $|\uparrow\downarrow\uparrow\downarrow\downarrow\uparrow\uparrow\uparrow\uparrow\downarrow\downarrow\rangle$ (d). Simulations are performed in the absence of a h_z field to ensure genuine interaction dynamics are observed. h_x is set to 0.1 to continue allowing quantum dynamics to occur. The states are evolved for a total of 100 time units in order to best observe the outcome of allowing the system to naturally evolve. Figures (a) - (c) demonstrate the interaction dynamics of systems with uniform sized bubbles. We observe from this, real-time interaction dynamics as in each case bubbles of ranging sizes can be expected to form. In each case an immediate spike from $\lambda_n = 0$ is observed. This makes sense; other than in the initial state, there should be at least some small chance to detect each bubble size. Figure (d) begins with system containing various sized bubbles and takes the time evolution as $t = 2,500$. We observe that throughout this evolution the lowest size remains dominant in the system, and are unable to observe any large peaks in $n = 2$ or $n = 3$ bubbles.

bubble growth in the long time limit. However, this of course goes against Coleman's predictions that large enough bubbles would consume the system. Since our results mostly match stages 1 and 2, it would be interesting to include detection of exceptionally large bubbles and simulate the system at increasingly large timescales to see if these predictions hold. By the same method we produced fig. 5d, which demonstrates the time evolution of an initial state containing bubbles of *various* size $n \leq 3$, time evolved for $t = 2500$ (i.e, much longer). We observe that the smallest size $n = 1$ remains dominant throughout. This is most consistent with [2] and [3] as we observe the proposed limitations stopping larger bubbles from expanding, altogether disagreeing with Coleman. With our theoretical framework properly underlined, it is worth considering a more realistic scenario, to see how this phenomenon acts in an actualised system.



(a) (b)

Figure 6: (a) Lattice structure of Cobalt Niobate ions (Co^{2+}) in a zig-zag formation. Each diagonal neighbour is at an angle of 31° to each other. ?This ensures that each Ion will most strongly feel the effects of its nearest neighbour but will also experience some contribution from its next neighbour. (b) for systems of N = 11 ions, we compute the mass ratios of bound state excitations for a Hamiltonian with open boundary conditions, (blue) and periodic boundary conditions, (green) compared to theoretical predictions for E_8 (orange). Here we have set $h_x = J' = 1$, $h_z = 0.05$, $J_p = 0.2$ and $J_B = 0.1$. These were chosen because the critical point was found to occur at $h_x \approx 0$. We therefore orient the system near the critical point, and split vacuum states through h_z without altering the configuration too drastically.

6 Bound States and E_8 Symmetry.

A notable experiment by Coldea et al [69], involved neutron scattering in Cobalt Niobate ($CoNb_2O_6$). This is a material that, to a reasonable approximation, can be viewed as a (chain) quantum Ising model [70]. The particles in the system are arranged in a zig-zag formation (fig. 6a), hence the effects of each particle's *next* nearest neighbour are also felt. As a perturbed critical Ising model, the system exactly acts like an integrable quantum field theory [71]. The experiment involved increasing the transverse magnetic field past its critical point, until the system underwent a quantum phase transition into a paramagnetic state.

A major finding from this experiment was that in this configuration, there existed particles in bound states with mass ratios that demonstrated excitations suggestive of E_8 symmetry, a special type of Lie algebra symmetry [72], that had previously been conjectured by Zamolodchikov [73] to appear in the critical Ising model. Lie Groups, such as SO(n) and U(n) [74], are *representations* of various continuous symmetries, and appear often in particle physics [75] and are important tools in quantum field theory [76]. E_8 is the largest "simple" symmetry group discovered in mathematics [77], and has not yet been realized experimentally, hence the significance of this implication. Developing our theory of systems relating to the Ising model could potentially be crucial for a more complete theory of false vacuum decay. This is because the "scattering matrix" [78] (often referred to as an S-matrix) would be exactly solvable, mean-

ing scattering amplitudes of meson states would be controllable experimentally. Clearly then, $CoNb_2O_6$ and E_8 symmetry are therefore worth exploration for developing our framework of false vacuum decay in more actualized systems.

The Hamiltonian for $CoNb_2O_6$ is given as:

$$\begin{aligned}\hat{H} = & -J' \sum_n \sigma_n^z \sigma_{n+1}^z - h_x \sum_n \sigma_n^x \\ & - h_z \sum_n \sigma_n^z - J \sum_n \left(\sigma_n^x \sigma_{n+1}^x + \sigma_n^y \sigma_{n+1}^y\right) + J_B \sum_n \sigma_n^z \sigma_{n+2}^z\end{aligned} \tag{10}$$

where each term is as described in fig. 6a. Numerical simulations of the time evolution for this system were difficult to obtain, and various techniques had to be used to combat this. By taking account of the Lieb-Robinson velocity of the system (as discussed in section 2), the system could be initialised as a chain of only a few ions, and more sites could be added to the chain over time. The experiment also used "Time-evolving block decimation" [79, 80], an algorithm for simplifying the process of numerically calculating the time evolution by identifying the relevant "block" of a much larger Hilbert space. This method is very commonly used in weakly entangled systems, such as the Ising model.

We are able to compute the mass ratios of bound state excitations via exact diagonalisation of this Hamiltonian. By sweeping through alternating values of h_x, we are able to find the optimal parameters to achieve a near-critical point for this Hamiltonian. We then apply a weak perturbation of $h_z > 0$ to break the symmetry, and set $J_B > 0$ to include next nearest neighbour interactions. By coding the Hamiltonian eq. (10), we could then use Lanczo's algorithm [81] to extract the low lying eigenvalues, in order to compute the normalised excitation spectrum. We demonstrate our results in fig. 6b, considering both open boundary conditions, and periodic boundary conditions, compared to the first few E_8 symmetry values as predicted by Zamolodchikov [73]. Exact values obtained are provided in the table below. Although our obtained results do not exactly replicate E_8, the general trend is clearly observed. We also were able to show that a system with periodic boundary conditions will obtain values closer to those representative of E_8 symmetry than a system that uses open boundary conditions.

Generally, we observe much better accuracy for particles in the lower mass ratio. This is likely because of finite size effects, which are well known to be difficult to work with [82]. Larger states are therefore much more sensitive to the initial conditions of the system, and difficulty is experienced when attempting to fit larger particles to E_8 theory. Furthermore, since E_8 occurs at the thermodynamic limit, and is hence best observed for $N \to \infty$, larger system sizes would provide more accurate representations. The reason that at times there is no jump observed between mass ratios is because of the natural translational symmetry of employing periodic boundary conditions. This means that the momentum of any excited state necessarily needs to be conserved. Furthermore, every excitation has an exactly-degenerate pair with opposite momenta, hence the energy splitting between some neighbouring particles can be intrinsically small [83].

E_8 Theory:	1	1.618	1.989	2.4048	2.9563	3.2183	3.891	4.7835
PBC:	1	1.639	1.639	2.2442	2.8853	2.8853	3.241	3.2414
Difference:	0	(+) 0.021	0.35	0.1606	0.071	0.333	0.65	1.5421

7 Discussion.

Throughout section 4, we have provided data obtained from performing simulations on $N = 12$ Ising model systems with periodic boundary conditions. Analysis of a larger sized system would create a more accurate portrait of the phenomenon; however, this was outside of the computational scope of the project. From the observed peaks in $\lambda_n(t)$, we were able to verify several predictions and observations from other papers. Based on the parameters used for detection, we demonstrated that larger bubbles would require a more disordered system ($h_x > 0.5$) to be achieved. We also showed that smaller bubbles would still maintain dominance in this regime.

$$h_z = -2J/n \tag{11}$$

In section 5, we were then able to simulate the time evolution of these bubbles. We observed a clear fluctuation between bubble sizes within the system, indicating the necessity to consider bubble interactions when analysing the time evolution of a mixed state system. A key finding in this sense is that interactions between bubbles are a pivotal component when considering the false vacuum decay process. We were also able to show that smaller bubbles are more commonly detected in small time intervals, consistent with predictions by Coleman [7]. A possible next step would be to simulate bubbles larger than the normal regime, $n > 6$ and attempt to demonstrate that in this case, the bubble will continuously expand and increase the expectation value of larger sized bubbles. Performing simulations on two or three dimensional Ising models could also yield more descriptive results.

In section 6, we considered a more realistic parallel to the Ising model, ($CoNb_2O_6$) and were able to demonstrate a correlation between the excitation of bound states near the continuum limit and the specific mass ratios characteristic of the E_8 Lie algebra structure. Development of our theory through mathematical structures such as E_8 is pivotal to understanding more complex scenarios, and directly complements our ability to undertake experimental projects. This demonstrates the necessity for both theoretical and experimental effort in advancing physics.

8 Conclusion.

As an overall conclusion, through our programmed model of the Ising model and false vacuum decay process, we have shown that the theoretical data for both bubble nucleation of false vacuum bubbles and its subsequent interaction dynamics match that of experimental data. Further, our interpretations of the theory underlining the process match the general consensus developed by other papers.

A full discussion of the quantum field theoretical aspects of false vacuum decay lies beyond the scope of this study. However, we have been able to describe a multitude of physical scenarios through reference and understanding of the quantum Ising model. This demonstrates the effectiveness of the model in simplifying more complex scenarios through exploiting the continuous symmetry of QFTs. Symmetry is a key tool for quantum field theorists, and hence researching models with desirable symmetries (such as the Ising model), could be pivotal to solving the current issues surrounding QFT. Another way in which the theory could be developed would be through improvements to the measurement precision of the top quark [84], as this would give us a better understanding of the electroweak vacuum, an obvious parallel to the false vacuum.

Modern experiments are attempting to utilise an increasingly large number of entangled qubits, increasing the issue of decoherence. This has proven to be a significant limitation to the scale of simulations that are achievable. Clearly then, as proposed by Darbha et al [28] the main limiting factor in further research is simply computational. Larger system sizes ($N > 1,000$) could be pivotal in further breakthroughs, however this would also require a better understanding of how to keep systems from becoming decoherent [85, 86]. Advances in quantum computing and machine learning could therefore be key to a natural progression to studying more complicated systems, thereby gaining a deeper understanding of this phenomenon.

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